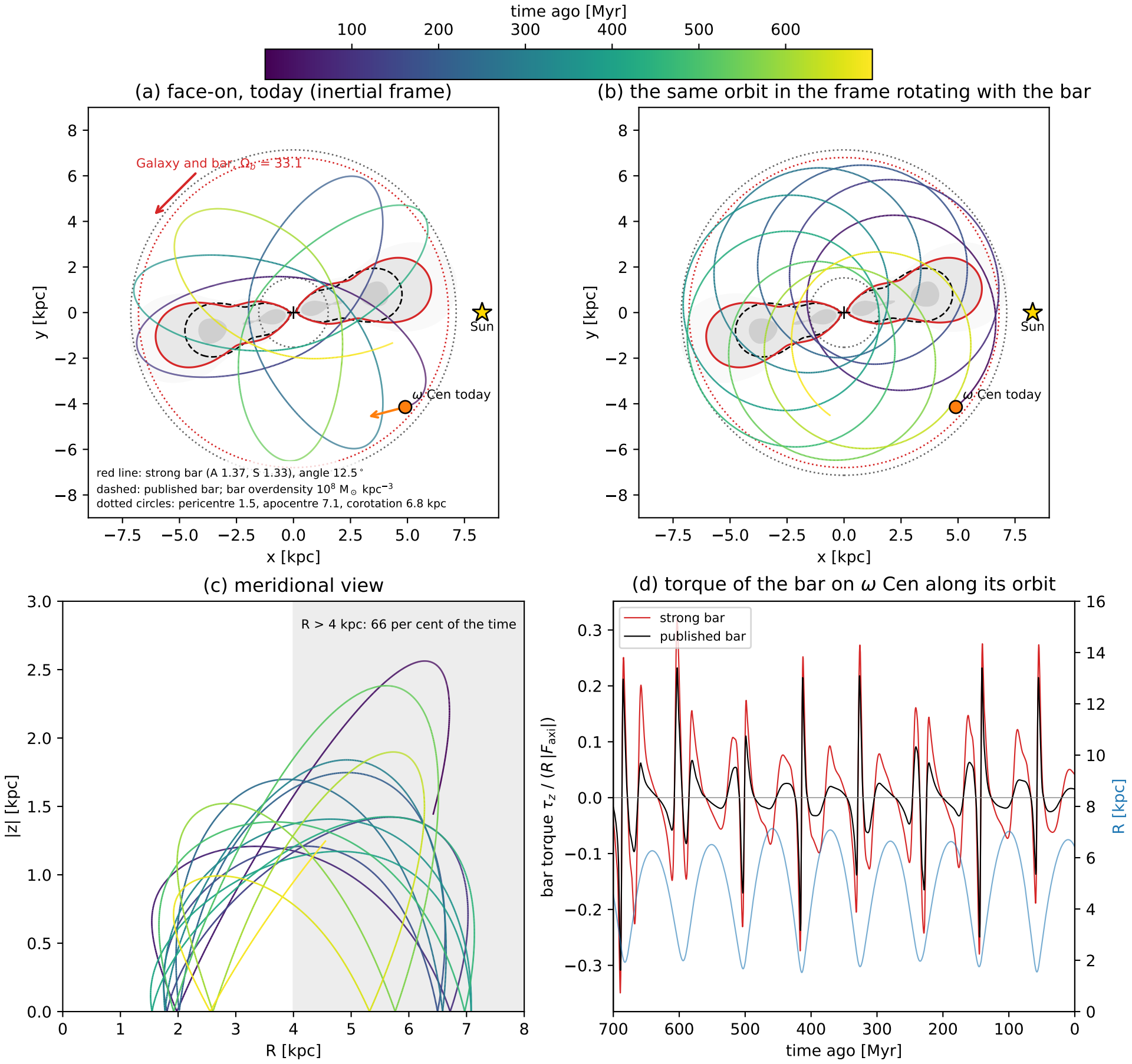

time ago [Myr]
100
200
300
400
500
600
(a) face-on, today (inertial frame)
(b) the same orbit in the frame rotating with the bar
Galaxy and bar, Ω_b = 33.1
Sun
ω Cen today
red line: strong bar (A 1.37, S 1.33), angle 12.5°
dashed: published bar; bar overdensity 10^8 $M_\odot$ kpc^{-3}
dotted circles: pericentre 1.5, apocentre 7.1, corotation 6.8 kpc
x [kpc]
y [kpc]
(c) meridional view
R > 4 kpc: 66 per cent of the time
R [kpc]
|z| [kpc]
(d) torque of the bar on ω Cen along its orbit
strong bar
published bar
bar torque τ_z / (R $|F_{axi}|$)
R [kpc]
time ago [Myr]